

AI Tools & Reflective Journal

CNN Foundation — Deep Learning Project

Michael Mizzi - [April – May 2026]

A. Tools Used

The following Generative AI and research tools were employed across the project lifecycle, from initial literature research through to simulator development and presentation preparation:

- **Research of CNN papers, consolidation of paper lists, concept explanation during paper review, study notes text building, image generation, and cross-evaluation of study notes.**
Tools: Google Gemini, Claude, Perplexity AI. — *Refer to Prompt #1*
- **Locating and verifying peer-reviewed links for each selected paper.**
Tool: Google Scholar [Labs]
- **Literature organisation, understanding technical details, infographic generation, mind maps, multiple-choice quiz drafting, data tables, and video summaries.**
Tool: NotebookLM — *Refer to Prompt #2*
- **Guidance to integrate Gradio / Streamlit to the Interactive simulator code and setting as a standalone app.**
Tool: Gemini / Claude
- **UI layout improvement of the simulator, and cross-evaluation of study notes.**
Tool: Claude (Anthropic) — *Refer to Prompt #3*
- **Conversion of quiz to Google Forms; layout enhancement of study notes**
Tool: Manus AI — *Refer to Prompt #4*
- **Drafting and iterating the CNN simulator; development of the walkthrough Jupyter Notebook for the worked example. Engaging MNIST dataset.**
Tool: Google Colab
- **Drafting the final presentation slide deck.**
Tool: Presentations.AI
- **Critical assessment, evaluation and scoring of the study notes pack.**
Tool: Gemini / Claude — *Refer to Prompt #5*

B. Reference Prompts

The following are examples of prompts that were used at key stages of the workflow:

Prompt 1 — Paper Research & Consolidation

Gemini	Claude
Describe in 5 bullet points summary the following: Block : CNN Foundations --> Convolution, Spatial Arithmetic and Receptive Fields Kernels, the output dimension equation, stride, padding, dilation, and effective receptive field CNN Foundations: Spatial Arithmetic & Receptive Fields Here is a 5-point breakdown of the mechanics that allow Convolutional Neural Networks to "see" and process spatial data:	Describe in 5 bullet points summary the following: Block : CNN Foundations --> Convolution, Spatial Arithmetic and Receptive Fields Kernels, the output dimension equation, stride, padding, dilation, and effective receptive field Synthesized CNN fundamentals across convolution, kernels, and receptive fields > • Convolution Operation: A convolution applies a small learnable matrix called a kernel (or filter) that slides over the input, computing dot products at each position to produce a feature map — this allows the network to detect local patterns (edges, textures) regardless of their position in the image.

list top 10 research papers related to this topic

Synthesized foundational CNN research papers from training knowledge >

Here are the top 10 foundational and influential research papers on this topic:

1. "Gradient-Based Learning Applied to Document Recognition" (1998) — LeCun et al.
The original LeNet paper that introduced CNNs for image recognition, establishing the convolution + pooling paradigm still used today.

2. "ImageNet Classification with Deep Convolutional Neural Networks" (2012) — Krizhevsky, Sutskever & Hinton. The AlexNet paper that reignited deep learning, demonstrating the power of stacked conv layers with stride and ReLU at scale.

3. "Very Deep Convolutional Networks for Large-Scale Image Recognition" (2014) — Simonyan & Zisserman (VGGNet). Systematically showed that using many small 3x3 kernels stacked deeply outperforms large kernels, directly informing kernel size intuition.

Output a csv file tabulating these 10 research papers and adding the URL to each

Show code >

Your CSV file containing the top 10 research papers related to CNN foundations, spatial arithmetic, and receptive fields is ready.

cnn_foundations_papers

CSV

list top 10 research papers related to this topic

To master the mechanics of CNNs, these 10 papers are the "canonical" texts. They range from the biological inspiration of receptive fields to the precise spatial arithmetic used in modern architectures.

1. The Biological Inspiration
"Receptive fields of single neurones in the cat's striate cortex" Hubel, D. H., & Wiesel, T. N. (1959) * Why it matters: The bedrock of the entire field. It discovered that neurons in the visual cortex respond to specific spatial patterns (edges/orientations), introducing the concept of the receptive field.

2. The First "Convolutional" Architecture
"Neocognitron: A self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position" Fukushima, K. (1980) * Why it matters: Introduced the S-layers (convolution) and C-layers (pooling), creating the structural template for modern CNNs to achieve shift-invariance. Wikipedia

3. The Modern Blueprint

Tabulate these research papers in a csv and list url

Orchestrated CSV compilation with paper details and URLs >

The CSV is ready with 6 columns — No., Title, Authors, Year, Topic, and URL. All URLs point to either the original publisher, arXiv preprint, or author-hosted PDFs for direct access.

Cnn research papers

Table: CSV

Open in Excel

From the above exercises, repeated as well using PerplexityAI we returned a consolidated spreadsheet with the top 10 papers that were listed by the three systems as relevant to the subject. Top 5-6 papers were then manually selected to base the study pack.

Prompt 2 — NotebookLM – understanding research papers / topic

[3-examples]

Saturday, 9 May

what are ANNs and why do they pose limitations for computer vision as compared to CNNs. Explain using a monkey-banana analogy

Artificial Neural Networks (ANNs), specifically standard multilayer feedforward networks, are systems where each unit in a hidden layer is fully connected to every single unit in the previous layer 1 2 . While they can learn complex, high-dimensional mappings, they pose three significant limitations for computer vision compared to Convolutional Neural Networks (CNNs):

describe 'padding' and explain types of padding used in CNNs

In Convolutional Neural Networks (CNNs), padding is the process of adding extra elements—most commonly zeros—to the borders of an input feature map before a convolution operation is performed 1 2 . This technique is used to control the spatial dimensions of the output and ensure that features at the edges of the input are processed appropriately 3 4 .

Purpose of Padding

- Controlling Output Size: Without padding, the spatial dimensions of a feature map naturally shrink after each convolutional layer because the kernel cannot be centered on the very edge pixels without falling off the boundary 5 6 .
- Border Feature Preservation: Padding allows the kernel to visit pixels at the extreme edges of the image as many times as central pixels, preventing the loss of information at the borders 4 7 .

explain convolution arithmetic

Convolution arithmetic provides the mathematical framework for determining the spatial dimensions of feature maps and the extent of receptive fields as data passes through the layers of a convolutional neural network (CNN) 1 2 . Unlike fully connected layers, where output size is independent of input size, a convolutional layer's output is governed by the interaction of several hyperparameters 3 .

1. The Forward Convolution Equation

The fundamental relationship used to calculate the output size (o) of a convolutional layer along any given axis, based on the input size (i), kernel size (k), padding (p), and stride (s), is:
$$o = \lfloor \frac{i + 2p - k}{s} \rfloor + 1$$
 4 .

The floor function (⌊ ⌋) accounts for cases where the stride does not allow the kernel to perfectly align with the end of the padded input, meaning that a few pixels are effectively ignored 5 6 .

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Prompt 3 — Simulator Development / UI Improvement

[Claude]

Redesigning Gradio CNN simulator layout with dark blue theme ▾



Convolution
demo_Gradio
pipeline.txt
264 lines

TXT

Background:

- 1) attached text file contains python code to build an interactive simulator using Gradio
- 2) image name 'current output' shows output based on code developed in 1)
- 3) image names 'recommeneded output' shows the desired output layout. note that each section is numbered identical to the numbers used in the 'current output' image

Required: update python code to change visible layout of simulator as close as possible to the 'recommeneded output' and using a dark blue colour scheme. Do not change logic of the CNN computations, kernels, padding, stride and dilations

Excellent updates that avoided all coding steps required to change Python and CSS code. The code was loaded as TXT purposely to ensure that manual verification is carried out prior to running. This avoids connecting directly to VSC and losing vision.

Prompt 4 — Quiz conversion from typed document to interactive Google forms

[Manus-AI]

Manus 1.6 Lite ▾

Upgrade

Share

📄

📄



quiz_with_rationale.docx
Document - 45.00 KB

convert this quiz into google forms and provide access link to share with people to take this quiz. Add interactive mode to show answer feedback as listed in document

manus

5/6

Got it — I'll review the quiz document and work on converting it into a shareable Google Form with answer

The resulting google forms template was fully functional with excellent integration. The weak side was the embedding of mathematical formula which had to be fine-tuned manually.

Prompt 5 — Study Notes Cross-Evaluation

[Claude] – Note: Same exercise carried out via Gemini



Assess the attached study notes about basics of 'convolution neural networks'. Evaluate the following aspects; 1) integrity of the different sections within this document 2) accuracy of the write-up 3) integrity and relevance of the referenced papers 4) easy of use of this document 5) importance and relevance of the images and schematics used. for each of these 5 aspects rank between 1 and 3 (1- not good, 2- average, 3-very good) and include a maximum of 3 bullet points to justify the ranking

The scoring by Gemini and Claude was consistent and both highlighted room for improvement on how the worked example was initially depicted. This was later modified to improve the explainability.

C. Reflective Questions

i) Which concepts did AI help explain?

Generative AI proved particularly useful in demystifying the theory of Convolutional Neural Networks. NotebookLM was used interactively during the manual review of the top six selected papers to clarify concepts such as convolutional operations and feature map construction. The mathematics was quite simple however the compounded effect of the hyperparameters are quite challenging to explain, an area where GenAI was fundamental in aiding with fast reproduced infographics/ visuals.

ii) Where did Gen AI return wrong information?

A notable issue arose with NotebookLM's initial handling of the LeCun 1998 paper: the tool failed to surface any content from it in its outputs. This was later identified as a text-decoding problem — the PDF had not been properly parsed, rendering the foundational paper invisible to the model. This represents a subtle but significant form of misinformation by omission. Additionally, early paper lists generated by AI tools (returning 40+ results) lacked the precision required for a focused literature review, suggesting that broad AI-generated searches should always be validated against authoritative academic sources such as Google Scholar.

iii) When did Gen AI waste your time?

An initial attempt to use Manus to enhance the layout of the study notes and produce a cheat sheet was noted as ineffective, requiring manual correction and ultimately yielding unsatisfactory results. Similarly, the initial over-reliance on broad AI-generated paper searches (40+ papers from NotebookLM web search) required significant manual filtering before a workable shortlist could be established. The early exploration of Streamlit for the simulator UI, pursued with Gemini's guidance, was later abandoned in favour of Gradio, representing an avoidable iteration cycle had more targeted tooling decisions been made upfront.

iv) When and where did Gen AI excel?

Gen AI delivered the most value during content synthesis and iterative refinement. NotebookLM, once supplied with correctly parsed papers and guided by custom prompts, represented a step change in quality across infographics, slides, quizzes, and data tables. Gemini excelled in real-time concept clarification during paper review sessions and in generating supporting images for the study notes. Claude contributed effectively to UI layout refinements for the simulator and to cross-evaluation of study notes alongside Gemini. The use of Google Colab with AI assistance for MNIST dataset integration and the development of the walkthrough Jupyter Notebook represented a smooth and productive AI-augmented development cycle.

v) Do you see any gaps in the explanations of the CNN foundation subject?

One potential gap revolves around the completeness of the CNN pipeline. Since this study pack, simulator, quiz and walkthrough notebook address only the first building blocks of a CNN, readers will not see a complete CNN pipeline end-to-end. In other words, do not expect to see an outcome of the CNN [classification/ detection/ segmentation]. Furthermore, the study notes delve into the interpretation of receptive field and the effective-receptive field [ERF] and yet most examples provided contain only a single layer of convolution. Thus, it fails to express the impact of multiple layers on the ERF. This is partially addressed in the walkthrough Jupyter notebook.

